

# Ice Rheology Part 2: Elasticity and Viscoelasticity

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## Topics Covered

- Linear elasticity and Hooke's law for isotropic materials
- Elastic moduli and wave speeds in ice
- Maxwell viscoelasticity and the relaxation time
- Elastic flexure of floating ice
- Examples: grounding zone tidal flexure, jökulhlaups, and supraglacial lake drainage

**Primary reference:** Cuffey & Paterson, *Physics of Glaciers*, 4th ed., Chapters 3 and 10; Petrenko & Whitworth, *Physics of Ice*, Chapter 4.

## 1 Introduction and Motivation

In the preceding lectures on ice rheology, we treated ice as a viscous fluid where stress determines the rate of deformation, and there is no memory of past states (or strain). Glen's flow law and the composite flow law of [1] describe the long-term, steady-state creep behavior of ice under sustained loading. This viscous description is appropriate for processes that unfold over timescales much longer than the material's relaxation time, a material property we define later that varies between hours and days in terrestrial glaciers.

Many important glaciological processes, however, occur on timescales short enough that ice might behave elastically. Tidal loading at grounding zones flexes ice shelves with periods of hours. Jökulhlaups drain subglacial lakes over hours to days. Supraglacial lakes load ice shelves and can drain catastrophically in minutes. On these timescales, ice can deform like an elastic solid where the Cauchy stress is proportional to strain (symmetric component of the gradient of displacement). In an elastic material, deformation is recoverable when the load is removed.

The goal of this lecture is to develop the elastic and viscoelastic constitutive relations for ice, building on the viscous framework established earlier. We begin with linear elasticity—the simplest constitutive model for a solid material—and derive Hooke's law. We then introduce the Maxwell viscoelastic model, which combines elastic and viscous behavior in a single framework and provides a natural transition between the short-timescale elastic response and the long-timescale viscous flow we have already studied. Worked examples illustrate the application of these ideas to grounding zone flexure, jökulhlaups, and ice-shelf hydrofracture.

Two appendices derive Hooke's law and the heat equation from the first law of thermodynamics, showing how elasticity (the reversible storage of energy through molecular displacement) and heat

conduction (the irreversible transport of energy through molecular vibration) arise from the same fundamental energy balance.

**Plan for this lecture.** We review linear elasticity and Hooke’s law (§2), present the elastic properties of ice (§3), develop the elastic flexure equation for thin plates (§4), introduce Maxwell viscoelasticity (§5), and work through three glaciological examples (§6). Appendix A derives Hooke’s law from the first law of thermodynamics; Appendix B derives the standard heat equation; and Appendix C derives the generalized heat equation that includes viscous dissipation.

## 2 Linear Elasticity

### 2.1 Stress and Infinitesimal Strain

In the viscous framework developed in earlier lectures, the constitutive relation links stress to the *rate* of deformation:  $\dot{\varepsilon}_{ij} = f(\tau_{ij})$ . In an elastic solid, the constitutive relation links stress to the deformation itself:  $\sigma_{ij} = g(\varepsilon_{ij})$ .

The Cauchy stress tensor  $\sigma_{ij}$  and its decomposition into pressure and deviatoric parts,

$$\sigma_{ij} = -p\delta_{ij} + \tau_{ij}, \quad p = -\frac{1}{3}\sigma_{kk}, \quad \tau_{kk} = 0, \quad (1)$$

are the same as in the viscous theory. What changes is the kinematic quantity to which stress is related.

For small deformations, we define the **infinitesimal strain tensor**:

$$\varepsilon_{ij} = \frac{1}{2} \left( \frac{\partial s_i}{\partial x_j} + \frac{\partial s_j}{\partial x_i} \right), \quad (2)$$

where  $s_i$  is the displacement of a material point from its reference position. We reserve  $u_i$  for velocity, as in the viscous theory, so that  $u_i = \dot{s}_i$  and the strain rate is  $\dot{\varepsilon}_{ij} = \frac{1}{2}(\partial u_i / \partial x_j + \partial u_j / \partial x_i)$ . The two frameworks are connected by a time derivative:  $\dot{\varepsilon}_{ij}$  is the rate of change of  $\varepsilon_{ij}$ . Strain  $\varepsilon_{ij}$  is always defined relative to the reference (zero-stress) state, whereas strain rate is the instantaneous rate of change of strain. Because strain is a state variable, it evolves over time and not necessarily at the same rate nor even monotonically.

The “small deformation” assumption means  $|\varepsilon_{ij}| \ll 1$ , allowing us to drop higher-order terms in the full expansion and retain only the linear terms — hence, *linear* elasticity. In glaciology, elastic strains are tiny: a stress of  $\tau_e = 0.1$  MPa produces an elastic strain of order  $\varepsilon \sim \tau_e / \mu \sim 10^5 / 3.5 \times 10^9 \approx 3 \times 10^{-5}$ . The small-strain approximation is therefore excellent for the elastic response of ice. (Pedantic: Linear elasticity strictly requires both small strains and a stress–strain relation that is linear and rate-independent. Ice exhibits a weak frequency dependence of its elastic moduli—a form of anelasticity—which means the moduli measured at seismic frequencies differ slightly from those at tidal frequencies. This is a departure from ideal linear elasticity, though it is a small effect at glaciological stress levels.)

### 2.2 Hooke’s Law

For a linear elastic material, stress is a linear function of strain. The most general linear relation between two second-order tensors is

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl}, \quad (3)$$

where  $C_{ijkl}$  is the **fourth-order elastic stiffness tensor**. This is Hooke's law in its most general form. The stiffness tensor has  $3^4 = 81$  components, but symmetry reduces the number of independent components:

- Symmetry of the stress tensor ( $\sigma_{ij} = \sigma_{ji}$ ) requires  $C_{ijkl} = C_{jikl}$ , reducing 81 to 54.
- Symmetry of the strain tensor ( $\varepsilon_{kl} = \varepsilon_{lk}$ ) requires  $C_{ijkl} = C_{ijlk}$ , reducing 54 to 36.
- The existence of a strain energy function (Appendix A) requires  $C_{ijkl} = C_{klij}$ , reducing 36 to 21.

For a fully anisotropic crystal, 21 independent elastic constants are needed. Ice Ih has hexagonal symmetry, which reduces this to 5 independent constants. For polycrystalline ice with randomly oriented grains—the isotropic approximation—only 2 independent constants remain.

### 2.3 Isotropic Hooke's Law

For an **isotropic** material, the stiffness tensor must be invariant under all rotations. The most general fourth-order isotropic tensor with the required symmetries is

$$C_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad (4)$$

where  $\lambda$  and  $\mu$  are the **Lamé parameters**. Substituting into equation (3):

$$\boxed{\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij}.} \quad (5)$$

This is Hooke's law for an isotropic solid. The parameter  $\mu$  is the **shear modulus** (also denoted  $G$ ), which relates shear stress to shear strain. The parameter  $\lambda$  is the **first Lamé parameter**, which does not have as clean a physical interpretation on its own but appears naturally in the isotropic form.

The volumetric and deviatoric parts of Hooke's law separate cleanly. Taking the trace of equation (5):

$$\sigma_{kk} = (3\lambda + 2\mu) \varepsilon_{kk}, \quad (6)$$

so that  $p = -\frac{1}{3}(3\lambda + 2\mu) \varepsilon_{kk}$ . The **bulk modulus**  $K = \lambda + \frac{2}{3}\mu$  controls the volumetric response. The deviatoric part is

$$\tau_{ij} = 2\mu e_{ij}, \quad (7)$$

where  $e_{ij} = \varepsilon_{ij} - \frac{1}{3}\varepsilon_{kk} \delta_{ij}$  is the deviatoric strain. The shear modulus  $\mu$  alone controls the deviatoric (shape-changing) response.

### Key point: two independent moduli

An isotropic elastic solid is fully characterized by two independent elastic constants. Common pairs include:

| Pair           | Physical interpretation                                                        |
|----------------|--------------------------------------------------------------------------------|
| $\lambda, \mu$ | Lamé parameters (natural in the tensor form)                                   |
| $K, \mu$       | Bulk modulus (volumetric) and shear modulus (deviatoric)                       |
| $E, \nu$       | Young's modulus (uniaxial stiffness) and Poisson's ratio (lateral contraction) |

All pairs are equivalent; the choice is one of convenience. The relations are:

$$\lambda = K - \frac{2}{3}\mu = \frac{E\nu}{(1+\nu)(1-2\nu)}, \quad \mu = \frac{E}{2(1+\nu)}, \quad K = \frac{E}{3(1-2\nu)}. \quad (8)$$

## 3 Elastic Properties of Ice

### 3.1 Measured Elastic Moduli

The elastic moduli of polycrystalline ice Ih have been measured by a variety of techniques, including ultrasonic wave propagation, Brillouin spectroscopy, and mechanical testing. The most precise values come from wave-speed measurements, which probe the high-frequency (truly elastic) response without contamination from viscous creep. Table 1 summarizes the isotropic elastic constants for polycrystalline ice near  $-10^\circ\text{C}$  [2].

Table 1: Isotropic elastic moduli of polycrystalline ice Ih at  $-10^\circ\text{C}$  [2]. Values are Hill averages of the five single-crystal elastic constants, which provide the best estimate of the isotropic elastic response of a polycrystal with randomly oriented grains.

| Quantity             | Symbol    | Value    |
|----------------------|-----------|----------|
| Young's modulus      | $E$       | 9.33 GPa |
| Shear modulus        | $\mu$     | 3.52 GPa |
| Bulk modulus         | $K$       | 8.90 GPa |
| First Lamé parameter | $\lambda$ | 6.55 GPa |
| Poisson's ratio      | $\nu$     | 0.325    |

Several features of these values are worth noting:

1. **Ice is a stiff solid.** The shear modulus of ice ( $\mu = 3.5$  GPa) is about 25 times that of lead (0.14 GPa) and roughly 1/20 that of steel (80 GPa). The elastic response of ice is well described by the theory of linear elasticity—there is no measurable nonlinearity at glaciological stress levels.
2. **Poisson's ratio is close to 1/3.** A Poisson's ratio  $\nu = 0.325$  means that ice contracts laterally by about one-third of the amount it extends axially under uniaxial tension. For comparison,  $\nu = 0.5$  corresponds to an incompressible material (no volume change under any deformation), and  $\nu = 0$  corresponds to a material that does not contract laterally at all (like cork).

3. **Temperature dependence is weak.** The elastic moduli increase by roughly 5% between 0°C and −40°C [2]. This is a minor effect compared to the strong temperature dependence of viscous creep (activation energies of 49–180 kJ mol<sup>−1</sup>). For most glaciological calculations, the elastic moduli can be treated as constants.

### 3.2 Elastic Wave Speeds

The elastic moduli determine the speeds of seismic waves in ice. Substituting Hooke’s law (5) into Cauchy’s equation of motion ( $\sigma_{ij,j} + f_i = \rho \ddot{s}_i$ , in the absence of body forces) yields the Navier–Cauchy equation:

$$(\lambda + \mu) \frac{\partial^2 s_j}{\partial x_i \partial x_j} + \mu \frac{\partial^2 s_i}{\partial x_j \partial x_j} = \rho \frac{\partial^2 s_i}{\partial t^2}. \quad (9)$$

This wave equation admits two types of plane-wave solutions:

- **P-waves** (compressional):  $v_p = \sqrt{(\lambda + 2\mu)/\rho} = \sqrt{(K + \frac{4}{3}\mu)/\rho}$ ,
- **S-waves** (shear):  $v_s = \sqrt{\mu/\rho}$ .

For ice ( $\rho_i = 917 \text{ kg m}^{-3}$ ):

$$v_p = \sqrt{\frac{6550 + 2 \times 3520}{917}} \times 10^6 \approx 3850 \text{ m s}^{-1}, \quad v_s = \sqrt{\frac{3520 \times 10^6}{917}} \approx 1960 \text{ m s}^{-1}. \quad (10)$$

These values agree well with seismic observations in glaciers and ice sheets [3]. The P-wave speed in ice is notably higher than in water (1480 m s<sup>−1</sup>) and lower than in most rocks (4000–7000 m s<sup>−1</sup>).

## 4 Elastic Flexure of Thin Plates

Many glaciological applications of elasticity involve the bending of ice shelves and glacier tongues—structures that are thin compared to their horizontal extent. The theory of thin elastic plates provides an efficient framework for these problems.

### 4.1 The Flexure Equation

Consider a thin plate of thickness  $h$  and flexural rigidity

$$D = \frac{E h^3}{12(1 - \nu^2)}, \quad (11)$$

resting on or floating in a fluid of density  $\rho_w$ . Let  $w(x, y)$  denote the vertical deflection of the plate from its equilibrium position. For small deflections ( $|w| \ll h$ ), the governing equation is

$$\boxed{D \nabla^4 w + \rho_w g w = q(x, y)}, \quad (12)$$

where  $\nabla^4 = \partial^4/\partial x^4 + 2\partial^4/(\partial x^2 \partial y^2) + \partial^4/\partial y^4$  is the biharmonic operator,  $\rho_w g w$  is the buoyancy restoring force per unit area (for a floating plate), and  $q(x, y)$  is any applied load per unit area.

In one horizontal dimension (e.g., a beam or a plate that varies only in  $x$ ):

$$D \frac{d^4 w}{dx^4} + \rho_w g w = q(x). \quad (13)$$

The homogeneous solution (with  $q = 0$ ) involves a characteristic length scale, the **flexural wavelength**:

$$\ell_f = \left( \frac{4D}{\rho_w g} \right)^{1/4} = \left( \frac{E h^3}{3(1 - \nu^2) \rho_w g} \right)^{1/4}. \quad (14)$$

The flexural wavelength sets the horizontal distance over which the plate adjusts from a loaded to an unloaded state. Features much smaller than  $\ell_f$  are supported by the plate's rigidity; features much larger than  $\ell_f$  are in isostatic equilibrium with the underlying fluid.

## 4.2 Flexural Wavelength of Ice Shelves

For an ice shelf of thickness  $h$ , using  $E = 9.33$  GPa,  $\nu = 0.325$ , and  $\rho_w = 1025$  kg m<sup>-3</sup>:

$$\ell_f = \left( \frac{9.33 \times 10^9 \times h^3}{3 \times 0.894 \times 1025 \times 9.8} \right)^{1/4} = \left( \frac{9.33 \times 10^9}{2.70 \times 10^4} \right)^{1/4} h^{3/4} \approx 24.2 h^{3/4}, \quad (15)$$

where  $h$  is in meters and  $\ell_f$  is in meters. Representative values:

| Ice thickness $h$ (m) | Flexural rigidity $D$ (N m) | Flexural wavelength $\ell_f$ (km) |
|-----------------------|-----------------------------|-----------------------------------|
| 200                   | $6.9 \times 10^{15}$        | 1.3                               |
| 500                   | $1.1 \times 10^{17}$        | 2.6                               |
| 1000                  | $8.7 \times 10^{17}$        | 4.3                               |

The flexural wavelength is typically 1–5 km for Antarctic ice shelves. This length scale appears repeatedly in our worked examples.

## 4.3 Bending Stresses

The bending of a thin plate produces normal stresses that vary linearly through the thickness:

$$\sigma_{xx}(z) = -\frac{E z}{1 - \nu^2} \left( \frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right), \quad (16)$$

where  $z$  is measured from the plate's neutral surface (midplane), with  $z > 0$  upward. The maximum tensile stress occurs at the surface farthest from the center of curvature. For a plate deflecting downward ( $w < 0$ , concave upward), the maximum tensile stress is at the bottom surface ( $z = -h/2$ ). For a plate deflecting upward, the maximum tensile stress is at the top surface.

In one horizontal dimension:

$$\sigma_{xx}^{\max} = \frac{E h}{2(1 - \nu^2)} \left| \frac{d^2 w}{dx^2} \right| = \frac{6 |M|}{h^2}, \quad (17)$$

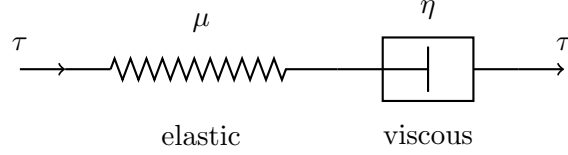
where  $M = -D d^2 w / dx^2$  is the bending moment per unit width.

# 5 Maxwell Viscoelasticity

## 5.1 The Maxwell Model

The creep flow laws developed in the preceding lecture describe the long-timescale behavior of ice under sustained loading. The elastic theory of §2 describes the instantaneous response to a

change in load. To bridge these regimes, we need a constitutive model that includes both elastic and viscous behavior. The simplest such model is the **Maxwell model**, which places an elastic element (spring) and a viscous element (dashpot) in series.



In a series arrangement, both elements carry the same stress, and the total strain rate is the sum of the individual strain rates:

$$\dot{\epsilon}_{ij} = \dot{\epsilon}_{ij}^{(\text{elastic})} + \dot{\epsilon}_{ij}^{(\text{viscous})}. \quad (18)$$

The elastic strain rate is the time derivative of the elastic constitutive relation (7):

$$\dot{\epsilon}_{ij}^{(\text{elastic})} = \frac{1}{2\mu} \dot{\tau}_{ij}. \quad (19)$$

The viscous strain rate is given by the creep flow law. For Glen's law:

$$\dot{\epsilon}_{ij}^{(\text{viscous})} = A \tau_e^{n-1} \tau_{ij}. \quad (20)$$

Combining:

$$\dot{\epsilon}_{ij} = \frac{1}{2\mu} \dot{\tau}_{ij} + A \tau_e^{n-1} \tau_{ij}. \quad (21)$$

This is the Maxwell viscoelastic constitutive equation for ice. The first term describes the instantaneous elastic response to changes in stress; the second describes the ongoing viscous creep under sustained stress.

Equation (21) is written for the deviatoric part of the stress-strain relation. For the volumetric part, we assume a purely elastic response:

$$\sigma_{kk} = 3K \epsilon_{kk}, \quad (22)$$

because ice does not undergo viscous volume change (the creep mechanisms are isochoric).

## 5.2 The Maxwell Relaxation Time

The Maxwell model has a characteristic timescale, the **relaxation time**, that separates elastic-dominated from viscous-dominated behavior. To find it, consider the 1D version of equation (21) and suppose the total strain is held fixed ( $\dot{\epsilon} = 0$ ) after an instantaneous loading. Then

$$\frac{1}{2\mu} \dot{\tau} + \frac{\tau}{2\eta} = 0, \quad (23)$$

where  $\eta$  is the effective viscosity, related to Glen's law by  $\eta = \tau_e / (2\dot{\epsilon}_e) = 1 / (2A \tau_e^{n-1})$ . For constant  $\eta$  (i.e., Newtonian viscosity or linearization about a given stress), the solution is exponential decay:

$$\tau(t) = \tau_0 \exp(-t/t_M), \quad t_M = \frac{\eta}{\mu}. \quad (24)$$

The **Maxwell time**  $t_M = \eta/\mu$  is the  $e$ -folding time for stress relaxation at constant strain. It has a direct physical meaning: elastic stresses decay on a timescale  $t_M$  as the viscous element takes up the deformation.

For ice with Glen’s law ( $n = 3$ ), the effective viscosity  $\eta = 1/(2A\tau_e^2)$  depends on the stress level, so the Maxwell time is also stress-dependent:

$$t_M = \frac{1}{2\mu A \tau_e^{n-1}}. \quad (25)$$

The Maxwell time is short where ice is warm and highly stressed, and long where ice is cold and under low stress. Table 2 gives representative values.

Table 2: Maxwell relaxation time  $t_M$  for polycrystalline ice under different conditions, computed from equation (25) using  $\mu = 3.5$  GPa,  $n = 3$ , and values of  $A$  from the Arrhenius relation with  $Q = 60$  kJ mol<sup>−1</sup> (below 263 K) and  $Q = 139$  kJ mol<sup>−1</sup> (above 263 K).

| Setting                   | $T$ (K) | $\tau_e$ (MPa) | $A$ (Pa <sup>−3</sup> s <sup>−1</sup> ) | $t_M$     |
|---------------------------|---------|----------------|-----------------------------------------|-----------|
| Warm shear margin         | 270     | 0.2            | $2.0 \times 10^{-24}$                   | 30 min    |
| Warm grounding zone       | 270     | 0.1            | $2.0 \times 10^{-24}$                   | 2.0 hours |
| Temperate glacier         | 263     | 0.1            | $3.5 \times 10^{-25}$                   | 11 hours  |
| Cold ice-sheet interior   | 243     | 0.05           | $1.2 \times 10^{-26}$                   | 55 days   |
| Cold, low-stress interior | 233     | 0.01           | $1.6 \times 10^{-27}$                   | ~28 years |

The Maxwell time ranges from about half an hour in warm, highly stressed ice (shear margins, grounding zones) to decades in the cold, low-stress interiors of ice sheets. This enormous range—more than five orders of magnitude—means that the same material can behave as an elastic solid or a viscous fluid depending on the local conditions and the timescale of the forcing.

### 5.3 The Deborah Number

A useful dimensionless number for deciding whether a process is elastic, viscous, or viscoelastic is the **Deborah number**:

$$\text{De} = \frac{t_M}{t_p}, \quad (26)$$

where  $t_p$  is the characteristic timescale of the process or forcing.

- $\text{De} \gg 1$ : the relaxation time is much longer than the process timescale. The viscous element has no time to respond, and the material behaves elastically.
- $\text{De} \ll 1$ : the relaxation time is much shorter than the process timescale. Elastic stresses relax away quickly, and the material behaves as a viscous fluid.
- $\text{De} \sim 1$ : both elastic and viscous effects are important, and the full Maxwell constitutive equation must be used.



### Key point: when does elasticity matter?

Elasticity matters when the Deborah number is of order unity or larger. For glacier ice, this means:

- **Tidal forcing** ( $t_p \sim 12$  hours):  $De \sim 0.04\text{--}0.9$  in warm ice near grounding zones. Viscoelastic effects are important.
- **Jökulhlaup drainage** ( $t_p \sim$  hours to days):  $De \sim 0.05\text{--}5$  depending on conditions. Both elastic and viscous responses are relevant.
- **Lake drainage / hydrofracture** ( $t_p \sim$  minutes to hours):  $De \gg 1$  everywhere. The response is essentially elastic.
- **Seismic waves** ( $t_p \sim 0.01\text{--}1$  s):  $De \gg 1$  everywhere. Purely elastic.
- **Seasonal flow variations** ( $t_p \sim$  months):  $De \ll 1$  in warm ice. Viscous.
- **Ice-sheet evolution** ( $t_p \sim 10^3\text{--}10^5$  years):  $De \ll 1$ . Purely viscous.

The line between “elastic” and “viscous” behavior is not a property of the material alone—it depends on both the material state (which controls  $t_M$ ) and the timescale of the process (which sets  $t_p$ ).

## 5.4 Response to Step Loading

The response of a Maxwell material to an instantaneous step load reveals the interplay between elastic and viscous deformation.

**Step stress (creep test).** Apply a constant stress  $\tau_0$  at  $t = 0$ . With  $\dot{\tau} = 0$  for  $t > 0$ , equation (21) gives:

$$\varepsilon(t) = \frac{\tau_0}{2\mu} + \frac{\tau_0}{2\eta} t = \frac{\tau_0}{2\mu} \left( 1 + \frac{t}{t_M} \right). \quad (27)$$

There is an instantaneous elastic strain  $\tau_0/(2\mu)$  at  $t = 0$ , followed by a steady viscous creep at rate  $\tau_0/(2\eta)$ . At  $t = t_M$ , the viscous strain equals the elastic strain. For  $t \gg t_M$ , the elastic contribution is negligible and the response is purely viscous.

**Step strain (relaxation test).** Apply an instantaneous strain  $\varepsilon_0$  at  $t = 0$  and hold it constant ( $\dot{\varepsilon} = 0$ ). From equation (24):

$$\tau(t) = 2\mu \varepsilon_0 \exp(-t/t_M). \quad (28)$$

The initial stress is  $2\mu \varepsilon_0$  (fully elastic), and it relaxes exponentially as the viscous element accommodates the imposed strain. After a time  $t_M$ , the stress has decayed to  $1/e \approx 37\%$  of its initial value. After  $3t_M$ , it has decayed to  $\sim 5\%$ .

**Oscillatory loading (harmonic forcing).** For a sinusoidal stress  $\tau(t) = \tau_0 \sin(\omega t)$ , the strain response of a linear Maxwell material is obtained by direct integration of the constitutive equation (21) (with  $\dot{\tau} = \tau_0 \omega \cos(\omega t)$  and  $\tau/(2\eta) = \tau_0 \sin(\omega t)/(2\eta)$ ):

$$\varepsilon(t) = \frac{\tau_0}{2\mu} \sin(\omega t) + \frac{\tau_0}{2\eta\omega} [1 - \cos(\omega t)]. \quad (29)$$

The first term is the elastic strain (in phase with the stress); the second is the viscous contribution, which oscillates about a mean offset of  $\tau_0/(2\eta\omega)$ . At high frequency ( $\omega t_M \gg 1$ ), the elastic

term dominates and the strain amplitude approaches  $\tau_0/(2\mu)$  (elastic limit). At low frequency ( $\omega t_M \ll 1$ ), the viscous term dominates and the strain amplitude scales as  $\tau_0/(\eta\omega) = \tau_0 t_M/(\mu)$ .

The tidal flexure of ice shelves near grounding zones is a natural example of oscillatory loading. The tidal period ( $\sim 12$  hours for the dominant  $M_2$  constituent) is comparable to the Maxwell time of warm ice near the grounding line, placing this problem squarely in the viscoelastic regime.

## 6 Worked Examples

### 6.1 Example 1: Grounding Zone Tidal Flexure

The grounding zone is the transition between grounded ice (resting on bedrock) and floating ice (an ice shelf or glacier tongue). Ocean tides cause the floating ice to rise and fall, while the grounded ice remains fixed. The transition between these two states involves flexure of the ice shelf over a horizontal distance comparable to the flexural wavelength.

**Setup.** Consider a one-dimensional cross section perpendicular to the grounding line. Let  $x$  denote distance from the grounding line, with  $x > 0$  in the floating ice shelf and  $x < 0$  in the grounded ice. The ice shelf has uniform thickness  $h$  and floats on the ocean with a tidally varying water level. The tidal displacement at  $x \rightarrow \infty$  is  $w_0(t) = w_0 \sin(\omega t)$ , where  $w_0$  is the tidal amplitude. At the grounding line ( $x = 0$ ), the ice is clamped:  $w(0, t) = 0$  and  $dw/dx|_{x=0} = 0$  (no deflection, no slope).

**Elastic solution.** The flexure equation (13) with  $q = 0$  has the general solution

$$w(x) = e^{-x/\ell_f} \left[ C_1 \cos\left(\frac{x}{\ell_f}\right) + C_2 \sin\left(\frac{x}{\ell_f}\right) \right] + e^{x/\ell_f} \left[ C_3 \cos\left(\frac{x}{\ell_f}\right) + C_4 \sin\left(\frac{x}{\ell_f}\right) \right], \quad (30)$$

where  $\ell_f = (4D/\rho_w g)^{1/4}$ . Requiring boundedness as  $x \rightarrow \infty$  sets  $C_3 = C_4 = 0$ . Far from the grounding line, the ice shelf is in hydrostatic equilibrium and moves with the tide:  $w(x \rightarrow \infty) = w_0$ . The solution satisfying the boundary conditions  $w(0) = 0$  and  $w'(0) = 0$  is

$$\boxed{w(x, t) = w_0 \sin(\omega t) \left[ 1 - e^{-x/\ell_f} \left( \cos \frac{x}{\ell_f} + \sin \frac{x}{\ell_f} \right) \right]}. \quad (31)$$

### Derivation: tidal flexure profile

The general bounded solution for  $x > 0$  is

$$w(x) = w_0 + e^{-x/\ell_f} \left[ C_1 \cos \frac{x}{\ell_f} + C_2 \sin \frac{x}{\ell_f} \right],$$

where we have added the far-field displacement  $w_0$  as a particular solution (the weight of the tide is supported by buoyancy, not by the plate). Applying  $w(0) = 0$ :  $C_1 = -w_0$ . For the slope condition, write  $\xi = x/\ell_f$ :

$$w(\xi) = w_0 + e^{-\xi} [C_1 \cos \xi + C_2 \sin \xi].$$

Differentiate:

$$w'(\xi) = e^{-\xi} [(-C_1 + C_2) \cos \xi + (-C_2 - C_1) \sin \xi].$$

At  $\xi = 0$ :  $w'(0) = -C_1 + C_2 = 0$ , so  $C_2 = C_1 = -w_0$ . Substituting:

$$w(x) = w_0 \left[ 1 - e^{-x/\ell_f} \left( \cos \frac{x}{\ell_f} + \sin \frac{x}{\ell_f} \right) \right]. \quad \square$$

**Bending stresses.** The maximum bending stress occurs at the grounding line ( $x = 0$ ), where the curvature is largest. From equation (31):

$$\left. \frac{d^2 w}{dx^2} \right|_{x=0} = \frac{2 w_0}{\ell_f^2}. \quad (32)$$

The maximum tensile stress at the ice surface is (from equation (17)):

$$\sigma_{\max} = \frac{E h}{2(1 - \nu^2)} \frac{2 w_0}{\ell_f^2} = \frac{E h w_0}{(1 - \nu^2) \ell_f^2}. \quad (33)$$

**Numerical evaluation.** For an ice shelf with  $h = 500$  m and a tidal amplitude  $w_0 = 1$  m:

- Flexural rigidity:  $D = 9.33 \times 10^9 \times 500^3 / [12 \times 0.894] = 1.09 \times 10^{17}$  N m.
- Flexural wavelength:  $\ell_f = (4 \times 1.09 \times 10^{17} / (1025 \times 9.8))^{1/4} = 2.6$  km.
- Maximum bending stress:  $\sigma_{\max} = 9.33 \times 10^9 \times 500 \times 1 / (0.894 \times (2600)^2) = 770$  kPa.

This stress is below but approaching the tensile strength of ice ( $\sim 1$  MPa). Tidal flexure alone does not typically fracture intact ice, but the stresses are large enough to promote fatigue crack growth in pre-existing crevasses at grounding zones, a process that has been linked to ice-shelf rifting.

**Viscoelastic correction.** The purely elastic solution (31) assumes that the ice responds instantaneously and completely to each tidal cycle. At the grounding zone of a warm ice shelf, the Maxwell time ( $t_M \sim 2$  hours; Table 2) is comparable to the tidal period ( $\sim 12$  hours for  $M_2$ ), so the Deborah number  $De \approx 0.17$ . The viscous component therefore accommodates a significant fraction of the tidal flexure. This has two consequences:

1. The effective flexural rigidity is reduced, because part of the deformation is viscous and unrecoverable. The flexural wavelength is shorter than the elastic prediction.

2. There is a phase lag between the tidal forcing and the ice-shelf response, because the viscous element introduces a delay.

Both effects are observed in GPS and tiltmeter measurements at Antarctic grounding zones [4, 5]. [9] used ICESat-2 laser altimetry to estimate the effective elastic modulus in grounding zones of the Ross Ice Shelf from tidal flexure profiles, finding that quantitative agreement with observed flexure requires accounting for viscoelastic effects through a reduced effective modulus.

## 6.2 Example 2: Jökulhlaups and Elastic Rebound

A jökulhlaup is a glacial outburst flood caused by the sudden drainage of a subglacial or ice-marginal lake. The drainage removes a distributed load from the glacier surface or bed, and the ice responds to this unloading. Whether the response is elastic, viscous, or viscoelastic depends on the drainage timescale relative to the Maxwell time.

**Setup.** Consider a subglacial lake of area  $\mathcal{A}$  and depth  $d_\ell$  beneath an ice sheet of thickness  $h$ . The lake supports a fraction of the overburden pressure: the water pressure at the ice–lake interface is  $p_w = \rho_w g d_\ell$ . When the lake drains on a timescale  $t_d$ , this hydraulic support is removed and the ice above the former lake must be supported entirely by the surrounding ice and bed.

**Elastic limit: fast drainage.** If  $t_d \ll t_M$ , the ice responds elastically. The removal of the water load  $q = \rho_w g d_\ell$  over the lake area produces an elastic subsidence of the ice surface. For a circular lake of radius  $R$ :

- If  $R \ll \ell_f$ : the plate is effectively rigid over the lake area, and the deflection is controlled by the plate’s flexural resistance. The maximum subsidence scales as  $w \sim q R^4/D$ .
- If  $R \gg \ell_f$ : the center of the former lake reaches isostatic equilibrium while the edges are supported by flexure.

For a more instructive estimate, consider the Deborah number. Suppose the ice above the lake is at  $T = 270$  K with  $\tau_e \approx 0.1$  MPa. From Table 2,  $t_M \approx 2$  hours.

**Case 1: Grímsvötn-type jökulhlaup** ( $t_d \sim 1$ – $2$  weeks). The Deborah number is  $De = t_M/t_d \approx 2/(24 \times 10) \approx 0.008$ . The drainage is slow compared to the Maxwell time, so the ice responds viscously. Elastic stresses relax as the lake drains, and the surface subsidence is controlled by viscous flow. GPS observations of surface displacement show smooth, gradual subsidence consistent with a viscous response.

**Case 2: Rapid subglacial lake drainage** ( $t_d \sim 2$  hours).  $De = t_M/t_d \approx 1.0$ . The drainage timescale is comparable to the Maxwell time. The initial surface response is elastic—a rapid subsidence of order centimeters to decimeters—followed by viscous relaxation over subsequent hours to days. Both elastic and viscous effects are important, and the full Maxwell model is needed to interpret surface displacement observations.

**Elastic subsidence estimate.** For an instantaneous drainage of a subglacial lake with  $d_\ell = 10$  m and  $R = 5$  km beneath  $h = 1000$  m of ice:

$$q = \rho_w g d_\ell = 1000 \times 9.8 \times 10 = 98 \text{ kPa}, \quad (34)$$

and the flexural wavelength is  $\ell_f \approx 4.3$  km (for  $h = 1000$  m, table in §4). Since  $R \approx \ell_f$ , the plate rigidity and buoyancy restoring force are both important. The maximum elastic deflection scales as

$$w_{\max} \sim \frac{q R^4}{D} = \frac{98 \times 10^3 \times (5000)^4}{8.7 \times 10^{17}} \approx 70 \text{ m}. \quad (35)$$

This crude scaling omits geometric prefactors that depend on the boundary conditions and load distribution. For a clamped circular plate, the central deflection is  $q R^4 / (64 D) \approx 1.1$  m; the exact value for a floating plate lies between this and the full scaling estimate. The order of magnitude—decimeters to a meter—is consistent with GPS observations of elastic rebound following rapid subglacial lake drainage events in Antarctica [6].

### 6.3 Example 3: Supraglacial Lake Drainage and Ice-Shelf Hydrofracture

Meltwater lakes that form on the surface of ice shelves can trigger a catastrophic chain of events: the weight of the lake water flexes the ice shelf, producing tensile stresses at the base. If these stresses are large enough, they can initiate a fracture that propagates vertically through the full ice-shelf thickness—a process called **hydrofracture**. Hydrofracture has been proposed as the mechanism that caused the rapid disintegration of the Larsen B Ice Shelf in 2002 [7, 8].

**Why elasticity?** Lake loading and unloading occur on timescales of days to weeks (filling) and minutes to hours (drainage). The Deborah number for the drainage phase is  $De \gg 1$ : the ice shelf responds elastically. The elastic analysis therefore governs whether hydrofracture can initiate.

**Setup.** Consider a circular supraglacial lake of radius  $R$  and depth  $d_w$  on an ice shelf of uniform thickness  $h$ . The lake applies a load  $q = \rho_w g d_w$  to the ice-shelf surface (taking the lake water density as  $\rho_w \approx 1000 \text{ kg m}^{-3}$ ). The ice shelf floats on the ocean, so the flexure equation (12) governs the elastic deflection.

**Flexural stress beneath the lake.** The lake load causes the ice shelf to deflect downward beneath the lake and upward at its edges. The downward deflection produces a concave-up curvature, which puts the bottom of the ice shelf in tension. For a lake much smaller than the flexural wavelength ( $R \ll \ell_f$ ), the maximum bending stress at the base of the ice shelf (at the center of the lake) scales as

$$\sigma_{\text{tens}} \sim \frac{3(1+\nu)\rho_w g d_w R^2}{8h^2}. \quad (36)$$

This is the maximum tensile stress at  $z = -h/2$  beneath a uniform circular load on a thin plate, valid for  $R \ll \ell_f$ . The factor  $(1+\nu)$  arises from the biaxial bending moment at the center of an axisymmetric load.

### Derivation: flexural stress beneath a circular lake load

For a uniform pressure  $q$  over a circle of radius  $R$  on an infinite plate floating on a fluid, the central deflection (for  $R \ll \ell_f$ ) is approximately

$$w_0 \approx \frac{q R^4}{64 D}.$$

The curvature at the center is  $\kappa = d^2 w / dr^2|_{r=0} \approx q R^2 / (16 D)$  (from the exact Kelvin function solution in the limit  $R/\ell_f \rightarrow 0$ ). At the center of an axisymmetric problem,  $\kappa_{rr} = \kappa_{\theta\theta} = \kappa$ , so the bending stress at the bottom surface ( $z = -h/2$ ) is

$$\sigma_{\text{tens}} = \frac{E h}{2(1 - \nu^2)} (1 + \nu) \kappa = \frac{E h (1 + \nu)}{2(1 - \nu^2)} \frac{q R^2}{16 D} = \frac{E h}{2(1 - \nu)} \frac{q R^2 12(1 - \nu^2)}{16 E h^3} = \frac{3(1 + \nu) q R^2}{8 h^2}.$$

Substituting  $q = \rho_w g d_w$ :

$$\sigma_{\text{tens}} \approx \frac{3(1 + \nu) \rho_w g d_w R^2}{8 h^2}. \quad \square$$

**Numerical evaluation.** Consider a lake with  $d_w = 2$  m and  $R = 500$  m on an ice shelf of thickness  $h = 200$  m. The flexural wavelength is  $\ell_f \approx 1.3$  km, so  $R/\ell_f \approx 0.4$  and the small-load approximation is reasonable. Using  $\nu = 0.325$ :

$$\sigma_{\text{tens}} \approx \frac{3 \times 1.325 \times 1000 \times 9.8 \times 2 \times 500^2}{8 \times 200^2} = \frac{1.95 \times 10^{10}}{3.2 \times 10^5} \approx 61 \text{ kPa}. \quad (37)$$

This is about 6% of the tensile strength of intact ice ( $\sim 1$  MPa). A deeper lake ( $d_w = 5$  m) or a larger lake ( $R = 1$  km) would produce stresses of 150 kPa or 240 kPa, respectively—still below the tensile strength of intact ice but potentially sufficient to reactivate pre-existing crevasses, which have a much lower critical stress intensity.

**The hydrofracture feedback.** Once a crevasse begins to propagate from the base (or surface) of the ice shelf, the water that enters the crack provides a pressure that drives the crack deeper. The key physics is that the hydrostatic pressure of water in the crack exceeds the lithostatic (ice overburden) pressure because  $\rho_w > \rho_i$ : at every depth, the water pressure in a water-filled crevasse exceeds the compressive stress from the weight of the overlying ice. This means a water-filled crevasse can, in principle, propagate through the full ice-shelf thickness regardless of thickness—a result first demonstrated by [10] and extended to ice shelves by [11, 7].

The elastic flexural analysis provides the initial stress state that determines whether hydrofracture can initiate. The subsequent propagation involves fracture mechanics (stress intensity factors, crack-tip processes) and the coupling between the fracture, the water supply, and the evolving stress field—topics beyond the scope of this lecture.

**Connection to ice-shelf collapse.** Prior to the disintegration of the Larsen B Ice Shelf in March 2002, satellite imagery showed extensive supraglacial lake formation on the shelf surface during the warm austral summer. Many of these lakes drained rapidly in the days immediately preceding the collapse. The elastic flexural stresses from the filling and draining of these lakes—compressive during filling, tensile during draining (from the elastic rebound)—are thought to have produced a “domino effect” in which the draining of one lake stressed neighboring regions, triggering further fracture and lake drainage [8]. This example illustrates how elastic processes operating on timescales

of hours to days can trigger ice-shelf disintegration events that reshape the ice-sheet system over decades.

## Reading

Cuffey & Paterson, *Physics of Glaciers*, 4th ed.:

- Section 3.2: Elasticity of ice
- Section 10.3: Flexure of floating ice
- Section 10.8: Calving and fracture

Petrenko & Whitworth, *Physics of Ice*, Chapter 4: Elastic properties.



## A Hooke's Law from the First Law of Thermodynamics

In this appendix, we derive Hooke's law starting from the first law of thermodynamics for a deformable body under the assumption of adiabatic conditions (no heat exchange with the surroundings). The derivation proceeds in five steps: (1) write the first law for an adiabatic process; (2) express the rate of work done on the body in terms of surface tractions and body forces; (3) apply the divergence theorem; (4) identify Cauchy's equation of motion (the Stokes equation for a viscous fluid, or Newton's second law for a continuum) in the resulting expression; (5) define the strain energy density and deduce the form of the stiffness tensor.

**Step 1: First law, adiabatic.** The first law of thermodynamics states that the rate of change of total energy (internal plus kinetic) equals the rate of work done by external forces plus the rate of heat added:

$$\frac{d}{dt}(U + \mathcal{K}) = \mathcal{P} + \mathcal{Q}, \quad (38)$$

where  $U$  is the total internal energy,  $\mathcal{K}$  is the kinetic energy,  $\mathcal{P}$  is the rate of work by external forces, and  $\mathcal{Q}$  is the rate of heat addition. For an adiabatic process,  $\mathcal{Q} = 0$ :

$$\frac{d}{dt}(U + \mathcal{K}) = \mathcal{P}. \quad (39)$$

**Step 2: Rate of work.** The external forces acting on a body occupying volume  $V$  with surface  $S$  are the surface tractions  $t_i$  and body forces  $f_i$  (per unit volume). The rate of work is

$$\mathcal{P} = \int_S t_i u_i dS + \int_V f_i u_i dV, \quad (40)$$

where  $u_i = \dot{s}_i$  is the velocity.

**Step 3: Divergence theorem.** Using Cauchy's traction formula  $t_i = \sigma_{ij} n_j$  (where  $n_j$  is the outward unit normal to  $S$ ), the surface integral becomes

$$\int_S \sigma_{ij} u_i n_j dS = \int_V \frac{\partial}{\partial x_j} (\sigma_{ij} u_i) dV = \int_V \left( \frac{\partial \sigma_{ij}}{\partial x_j} u_i + \sigma_{ij} \frac{\partial u_i}{\partial x_j} \right) dV, \quad (41)$$

where we applied the divergence theorem in the first equality and the product rule in the second.

**Step 4: Cauchy's equation of motion.** Substituting (41) into (40):

$$\mathcal{P} = \int_V \underbrace{\left( \frac{\partial \sigma_{ij}}{\partial x_j} + f_i \right)}_{= \rho \dot{u}_i} u_i dV + \int_V \sigma_{ij} \frac{\partial u_i}{\partial x_j} dV. \quad (42)$$

The bracketed term is Cauchy's equation of motion (conservation of linear momentum):

$$\frac{\partial \sigma_{ij}}{\partial x_j} + f_i = \rho \dot{u}_i. \quad (43)$$

This is the continuum form of Newton's second law. In the Stokes equation for viscous flow, the inertial term  $\rho \dot{u}_i$  is neglected (low Reynolds number), and the body force is gravity ( $f_i = \rho g_i$ ); the form is the same.

The first integral in (42) is the rate of change of kinetic energy:

$$\int_V \rho \dot{u}_i u_i dV = \frac{d}{dt} \int_V \frac{1}{2} \rho u_i u_i dV = \frac{d\mathcal{K}}{dt}. \quad (44)$$

The second integral involves  $\sigma_{ij} \partial u_i / \partial x_j$ . Because  $\sigma_{ij}$  is symmetric, only the symmetric part of  $\partial u_i / \partial x_j$  contributes:

$$\sigma_{ij} \frac{\partial u_i}{\partial x_j} = \sigma_{ij} \dot{\varepsilon}_{ij}, \quad (45)$$

where  $\dot{\varepsilon}_{ij} = \frac{1}{2}(\partial u_i / \partial x_j + \partial u_j / \partial x_i)$  is the strain rate tensor.

Substituting into the first law (39):

$$\frac{dU}{dt} = \int_V \sigma_{ij} \dot{\varepsilon}_{ij} dV. \quad (46)$$

In terms of the internal energy per unit volume  $\mathcal{U}$ :

$$\boxed{\dot{\mathcal{U}} = \sigma_{ij} \dot{\varepsilon}_{ij}}, \quad (47)$$

or equivalently,  $d\mathcal{U} = \sigma_{ij} d\varepsilon_{ij}$ . All of the mechanical work done on the body (minus the kinetic energy) is stored as internal energy—strain energy—in the material. This is the defining property of an elastic material: deformation is reversible and stores energy.

**Step 5: Strain energy function and the stiffness tensor.** Equation (47) implies that  $\mathcal{U}$  is a function of the strain:  $\mathcal{U} = \mathcal{U}(\varepsilon_{ij})$ , and

$$\sigma_{ij} = \frac{\partial \mathcal{U}}{\partial \varepsilon_{ij}}. \quad (48)$$

A material for which such a strain energy function exists is called **hyperelastic**. For small strains, expand  $\mathcal{U}$  as a Taylor series about the natural (stress-free) state  $\varepsilon_{ij} = 0$ :

$$\mathcal{U} = \underbrace{\mathcal{U}_0}_{=0} + \underbrace{\frac{\partial \mathcal{U}}{\partial \varepsilon_{ij}} \Big|_0}_{=\sigma_{ij}|_0=0} \varepsilon_{ij} + \frac{1}{2} \underbrace{\frac{\partial^2 \mathcal{U}}{\partial \varepsilon_{ij} \partial \varepsilon_{kl}} \Big|_0}_{=C_{ijkl}} \varepsilon_{ij} \varepsilon_{kl} + \cdots \quad (49)$$

The zeroth-order term is the energy of the reference state (set to zero). The first-order term vanishes because the reference state is stress-free. The leading nontrivial term is quadratic in strain, with coefficients

$$C_{ijkl} = \frac{\partial^2 \mathcal{U}}{\partial \varepsilon_{ij} \partial \varepsilon_{kl}} \Big|_0. \quad (50)$$

Differentiating:

$$\sigma_{ij} = \frac{\partial \mathcal{U}}{\partial \varepsilon_{ij}} = C_{ijkl} \varepsilon_{kl}. \quad (51)$$

This is Hooke's law (3). The stiffness tensor  $C_{ijkl}$  inherits the major symmetry  $C_{ijkl} = C_{klij}$  from the equality of mixed partial derivatives of  $\mathcal{U}$ —a consequence of the existence of the strain energy function, which in turn follows from the adiabatic first law.

**Isotropic reduction.** For an isotropic material,  $C_{ijkl}$  must be invariant under all proper orthogonal transformations (rotations). The most general isotropic fourth-order tensor with the symmetries  $C_{ijkl} = C_{jikl} = C_{ijlk} = C_{klij}$  is

$$C_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad (52)$$

giving

$$\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij}, \quad (53)$$

which is Hooke's law for an isotropic solid (5), with  $\lambda$  the first Lamé parameter and  $\mu$  the shear modulus.

The strain energy density for an isotropic solid is therefore

$$\mathcal{U} = \frac{1}{2}\lambda (\varepsilon_{kk})^2 + \mu \varepsilon_{ij} \varepsilon_{ij} = \frac{1}{2}K (\varepsilon_{kk})^2 + \mu e_{ij} e_{ij}, \quad (54)$$

where the second form separates the volumetric ( $K$ ) and deviatoric ( $\mu$ ) contributions. This confirms the physical interpretation:  $K$  controls the energy cost of volume change, and  $\mu$  controls the energy cost of shape change.

## B The Heat Equation from the First Law

In Appendix A, we assumed adiabatic conditions ( $\mathcal{Q} = 0$ ) and obtained the elastic constitutive relation. Here, we take the complementary limit: allow heat flow but neglect mechanical work ( $\mathcal{P} = 0$ , or more precisely, neglect the strain-energy contribution). This yields the standard heat equation.

**First law with heat flow.** With no mechanical work, the first law (38) reduces to

$$\frac{dU}{dt} = \mathcal{Q}. \quad (55)$$

The rate of heat addition to a material volume  $V$  has two contributions: conduction through the surface  $S$  and internal heat generation at rate  $H$  per unit mass:

$$\mathcal{Q} = - \int_S q_i n_i dS + \int_V \rho H dV = \int_V \left( -\frac{\partial q_i}{\partial x_i} + \rho H \right) dV, \quad (56)$$

where  $q_i$  is the heat flux vector (positive in the direction of heat flow, hence the negative sign: heat flowing *out* through the surface reduces the internal energy) and  $H$  includes radiogenic heating or other volumetric sources. We have applied the divergence theorem to convert the surface integral to a volume integral.

Writing the internal energy per unit volume as  $\mathcal{U} = \rho c_p (T - T_{\text{ref}})$  for a material with constant specific heat capacity  $c_p$ :

$$\rho c_p \frac{\partial T}{\partial t} = -\frac{\partial q_i}{\partial x_i} + \rho H. \quad (57)$$

**Fourier's law.** The constitutive relation for heat conduction is Fourier's law:

$$q_i = -k \frac{\partial T}{\partial x_i}, \quad (58)$$

where  $k$  is the thermal conductivity. For ice,  $k \approx 2.1 \text{ W m}^{-1} \text{ K}^{-1}$  near  $0^\circ\text{C}$  (and increases at lower temperatures). Substituting:

$$\boxed{\rho c_p \frac{\partial T}{\partial t} = k \nabla^2 T + \rho H.} \quad (59)$$

This is the **standard heat equation**—a parabolic partial differential equation that governs the diffusion of temperature through a material in the absence of mechanical coupling.

The thermal diffusivity  $\kappa = k/(\rho c_p)$  characterizes the rate of heat diffusion. For ice ( $\rho = 917 \text{ kg m}^{-3}$ ,  $c_p = 2090 \text{ J kg}^{-1} \text{ K}^{-1}$ ,  $k = 2.1 \text{ W m}^{-1} \text{ K}^{-1}$ ):

$$\kappa = \frac{2.1}{917 \times 2090} = 1.1 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}. \quad (60)$$

The characteristic diffusion time over a length scale  $L$  is  $t_{\text{diff}} \sim L^2/\kappa$ . For an ice thickness of  $h = 1000 \text{ m}$ :  $t_{\text{diff}} \sim 10^6/(1.1 \times 10^{-6}) \approx 9 \times 10^{11} \text{ s} \approx 29\,000 \text{ years}$ . Temperature changes in the interior of a thick ice sheet diffuse only over timescales comparable to glacial cycles.

## C The Generalized Heat Equation

Appendices A and B treated the two limits of the first law separately: adiabatic (no heat flow  $\rightarrow$  elasticity) and no work (no deformation  $\rightarrow$  heat equation). Here we retain both terms to obtain the **generalized heat equation**, which couples mechanical dissipation to the thermal field.

**Full first law.** Combining the results of the previous two appendices, the internal energy balance per unit volume is

$$\rho c_p \frac{\partial T}{\partial t} = \underbrace{\sigma_{ij} \dot{\epsilon}_{ij}}_{\text{mechanical work}} + \underbrace{k \nabla^2 T}_{\text{heat conduction}} + \rho H. \quad (61)$$

The mechanical work term  $\sigma_{ij} \dot{\epsilon}_{ij}$  represents the rate at which stresses do work on the deforming material. In the adiabatic elastic case, this work is stored as recoverable strain energy. In a viscous fluid, this work is dissipated irreversibly as heat.

**Viscous dissipation.** For an incompressible viscous material ( $\dot{\epsilon}_{kk} = 0$ , the condition appropriate for creeping glacier ice):

$$\sigma_{ij} \dot{\epsilon}_{ij} = (-p \delta_{ij} + \tau_{ij}) \dot{\epsilon}_{ij} = -p \dot{\epsilon}_{kk} + \tau_{ij} \dot{\epsilon}_{ij} = \tau_{ij} \dot{\epsilon}_{ij} \equiv \Phi, \quad (62)$$

where  $\Phi = \tau_{ij} \dot{\epsilon}_{ij}$  is the **viscous dissipation function**. Only the deviatoric stress does work; the pressure term vanishes because the flow is incompressible. Since both  $\tau_{ij}$  and  $\dot{\epsilon}_{ij}$  are related by the flow law,  $\Phi$  can be written in terms of the effective stress or effective strain rate alone:

$$\Phi = \tau_{ij} \dot{\epsilon}_{ij} = 2 \tau_e \dot{\epsilon}_e = 2A \tau_e^{n+1}, \quad (63)$$

where the second equality uses  $\dot{\epsilon}_e = A \tau_e^n$ . The dissipation is always positive ( $\Phi > 0$ ): viscous deformation converts mechanical energy irreversibly into heat.

Substituting into equation (61):

$$\boxed{\rho c_p \frac{\partial T}{\partial t} = 2A \tau_e^{n+1} + k \nabla^2 T + \rho H.} \quad (64)$$

This is the **generalized heat equation for glacier ice (assuming single-power-law viscosity)**. It states that the local temperature changes due to three effects: viscous dissipation (first term), thermal conduction (second term), and internal heat generation (third term).

**When does viscous dissipation matter?** The ratio of viscous heating to conductive heat transport scales as

$$\frac{\Phi}{k \nabla^2 T} \sim \frac{2A \tau_e^{n+1} h^2}{k \Delta T}, \quad (65)$$

where  $h$  is the ice thickness and  $\Delta T$  is the temperature difference across it. For typical ice-sheet conditions ( $\tau_e = 0.1$  MPa,  $A = 3.5 \times 10^{-25}$  Pa $^{-3}$  s $^{-1}$  at 263 K,  $h = 1000$  m,  $\Delta T = 30$  K,  $k = 2.1$  W m $^{-1}$  K $^{-1}$ ):

$$\frac{\Phi}{k \Delta T / h^2} = \frac{2 \times 3.5 \times 10^{-25} \times (10^5)^4 \times 10^6}{2.1 \times 30} = \frac{7 \times 10^{-25} \times 10^{26}}{63} = \frac{70}{63} \approx 1.1. \quad (66)$$

The two terms are comparable. Viscous dissipation is a significant heat source in the lower portions of thick, fast-flowing ice, where it warms the ice toward the pressure-melting point and softens it further—a positive feedback that concentrates deformation near the bed. In ice streams and shear

margins, where  $\tau_e$  is higher and the ice is warmer, viscous dissipation can be the dominant heat source.

**The connection between elasticity and heat.** The first law of thermodynamics provides a unified framework for understanding both elasticity and heat conduction. The adiabatic limit ( $\mathcal{Q} = 0$ ) yields elasticity: mechanical work is stored reversibly as potential energy through the displacement of molecules from their equilibrium positions. The no-work limit ( $\Phi = 0$ ) yields the heat equation: thermal energy, associated with the random vibrational motion of molecules, diffuses through the material. The full first law couples these two modes of energy: viscous dissipation converts ordered mechanical energy (the coherent translation and deformation of the material) into disordered thermal energy (the incoherent vibration of molecules about their lattice sites). Elasticity and heat are two manifestations of molecular-scale energy, distinguished by whether the molecular motion is ordered (elastic, recoverable) or disordered (thermal, irreversible).

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